

Project Report: FastAPI REST + PostgreSQL Blog API - Python Developer

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Family:	Python Developer
Level:	Intermediate
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Async FastAPI blog with SQLAlchemy 2.0, JWT auth, Alembic migrations, and Docker deployment.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Intermediate]

Chapter focus: Project Overview** *(Level: Intermediate)

FastAPI REST + PostgreSQL Blog API** is a intermediate-level end-to-end project for Python Developer. **Duration:** 3-4 weeks. Async FastAPI blog with SQLAlchemy 2.0, JWT auth, Alembic migrations, and Docker deployment. **Tools:** FastAPI, SQLAlchemy, Alembic, Docker.

Code Reference:

```
Objective: Async SQLAlchemy sessions
Objective: JWT login/register
Objective: CRUD posts with author ownership
Objective: Deploy on Railway/Docker
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Python Developer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Python Developer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Models [Intermediate]

Chapter focus: Phase: Models** *(Level: Intermediate)

User and Post SQLAlchemy models. Complete these tasks: Alembic migration; Relationships. Deliverable for this phase: Migrated schema.

Code Reference:

- [] Alembic migration
- [] Relationships

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Models is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Migrated schema to the teacher.

Chapter 3: Phase: Auth [Intermediate]

Chapter focus: Phase: Auth** *(Level: Intermediate)

OAuth2 password flow. Complete these tasks: JWT tokens; Protected routes. Deliverable for this phase: Auth Postman tests.

Code Reference:

- [] JWT tokens
- [] Protected routes

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Auth is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Auth Postman tests to the teacher.

Chapter 4: Phase: API [Intermediate]**Chapter focus: Phase: API** *(Level: Intermediate)**

Post CRUD with pagination. Complete these tasks: Pydantic v2 schemas; OpenAPI docs.
Deliverable for this phase: /docs working.

Code Reference:

- [] Pydantic v2 schemas
- [] OpenAPI docs

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase API is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the /docs working to the teacher.

Chapter 5: Phase: Deploy [Intermediate]**Chapter focus: Phase: Deploy** *(Level: Intermediate)**

Docker + CI. Complete these tasks: GitHub Actions; Railway deploy. Deliverable for this phase: Live URL.

Code Reference:

- [] GitHub Actions
- [] Railway deploy

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Deploy is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Live URL to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Intermediate]

Chapter focus: Final Deliverables and Review** *(Level: Intermediate)

Submit all final deliverables: Live API URL; OpenAPI spec; CI badge; Architecture README.
Skills demonstrated: FastAPI, Async, Auth, ORM. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Live API URL
Deliverable: OpenAPI spec
Deliverable: CI badge
Deliverable: Architecture README
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Live API URL and passes the rubric.